Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 October 2026
Team ID	Individual Project (No Team ID)
Project Name	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Connect to the provided global_fuel_prices_Dataset.csv file through Tableau Public. Use the available fields: date, country, region, income_level, subsidy_level, petrol_usd_liter, diesel_usd_liter, lpg_usd_liter, brent_crude_usd, and tax_percentage. No separate database or external live API is required for the current project.	High
2	Data Processing	Use Tableau data-source settings to confirm field data types and roles, use date as a time field, and use country, region, income level, and subsidy level as categories. Aggregate fuel prices, Brent crude price, and tax percentage for country, region, subsidy, income-level, and time comparisons. Any missing-value handling or data-cleaning steps should only be documented if actually performed.	High
3	Business Analysis	Calculate and compare average petrol price, average diesel price, average LPG price, and average tax percentage. Support country rankings, regional comparisons, taxation-versus-petrol-price analysis, subsidy-level comparisons, income-level comparisons, and time trends alongside Brent crude prices. Interpret associations as patterns, not proof of causation.	High
4	Business Insights & Analytical Outputs	Provide three interactive dashboards: (1) Fuel Taxation & Subsidy Policy, (2) Global Fuel Prices & Supply Chain Planning, and (3) Fuel Prices, Taxation & Economic Development. Include the project's country rankings, regional averages, taxation/price relationship views, subsidy comparisons, income-level comparisons, and historical trend view.	High
5	User Interaction & Analysis	Allow users to compare countries and regions, inspect the top and bottom 10 countries by average petrol price, view differences between petrol and diesel	High

		prices, and use the available Region legend and Income Level filters where included. Users should be able to explore patterns visually through dashboard charts and filters rather than edit source data.	
6	Data Update & Reporting	Provide access to the published Tableau Public workbook for viewing the dashboards. The current analysis uses the supplied CSV and its date coverage (January 2020–April 2026); it does not claim automatic or real-time updates. To refresh the analysis, the dataset would need to be updated and the workbook refreshed/published by the project owner.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboards should use clear titles, readable labels, consistent units (USD per litre and percentage), and straightforward comparisons so a student or general viewer can understand the displayed patterns. This is a design requirement, not a result from a formal usability test.
NFR-2	Security	The workbook is published on Tableau Public and should be treated as publicly viewable. Do not include private, confidential, or personally identifiable information. Access controls for a private enterprise deployment are outside the scope of this project.
NFR-3	Reliability	Charts and dashboard values should be based on the connected dataset and the workbook's defined aggregations. Verify that worksheets and dashboards load and display as intended before presenting or sharing. No formal reliability test has been claimed.
NFR-4	Performance	Dashboards should remain practical to navigate with the current dataset of approximately 27,468 records. Actual load times or performance benchmarks have not been measured, so no numeric performance guarantee is specified.
NFR-5	Availability	The workbook can be viewed when Tableau Public and the published workbook are accessible and the viewer has an internet connection. Availability is dependent on the external Tableau Public service; no uptime guarantee is claimed.
NFR-6	Scalability	The current workbook is designed for the supplied dataset and three dashboard scenarios. Larger datasets or more frequent updates may require data-source optimization, workbook redesign, or a different hosting/data architecture; this has not been load-tested.

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 October 2026
Team ID	Individual Project (No Team ID)
Project Name	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Tableau calculated fields and built-in aggregations	The project was developed in Tableau Public. Built-in calculations and aggregations support measures such as average petrol price and average tax percentage. No separate Python or SQL pipeline is documented as part of this project.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	CSV file: global_fuel_prices_Dataset.csv	The supplied CSV is the structured data source for the workbook. A separate database or data warehouse such as Snowflake or MySQL was not used.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau Public data connection to the CSV file	The dataset is connected to Tableau as a file-based data source, enabling the fields to be used in worksheets and dashboards.

4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau data-source settings, field roles/types, filters, and built-in aggregations	Analysis is performed in Tableau using the available fields, aggregations, and dashboard filters. This entry does not claim a separate ETL pipeline, SQL transformation, or external cleaning process.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau Public 2026.3 worksheets and dashboards	Tableau is used to create three dashboards covering taxation and subsidy policy, supply-chain price comparisons, and fuel prices by economic development indicators. Charts include rankings, regional comparisons, relationship plots, and time trends.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Published Tableau Public workbook	Users view and interact with the published workbook, including its dashboards, chart selections, and available filters such as income level. The published workbook is the user-facing interface.

Project Design Phase-II
Data Flow Diagram & User Stories

Date	31 October 2026
Team ID	Individual Project (No Team ID)
Project Name	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Maximum Marks	4 Marks

Data Flow Diagrams:

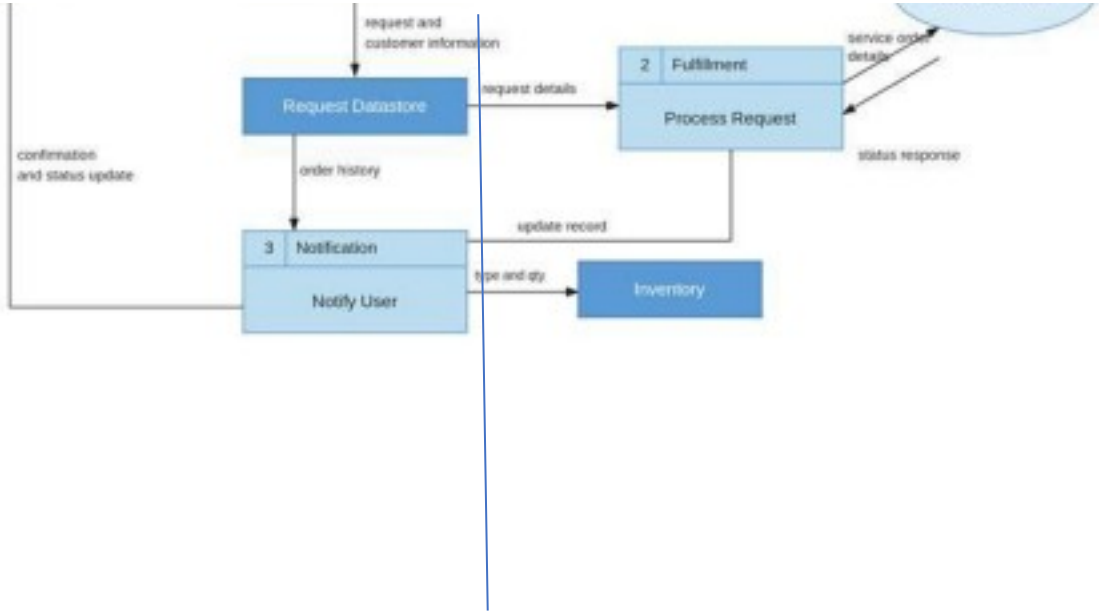
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).

Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example: DFD Level 0 (Industry Standard)



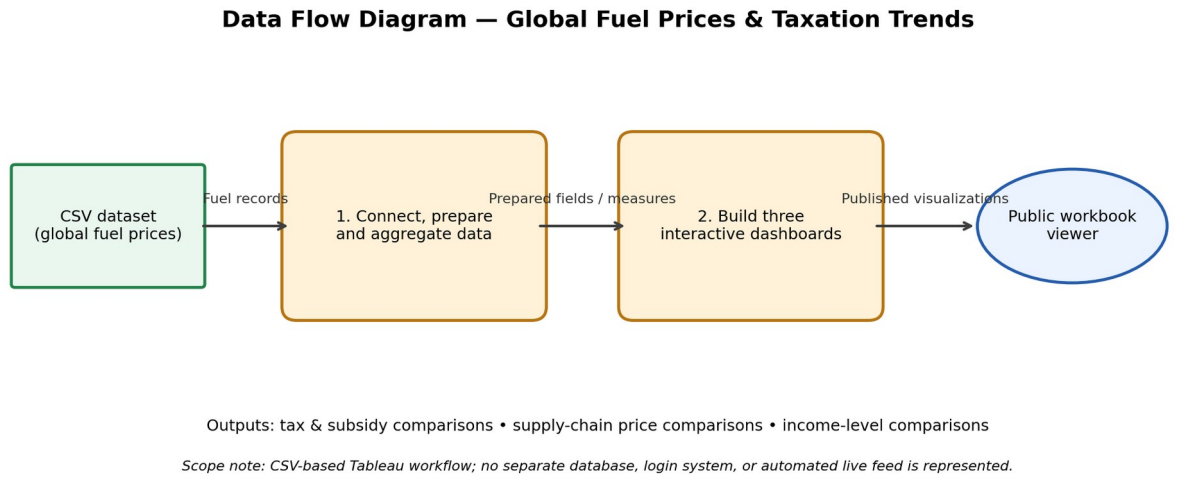


Figure: Project-level data flow based on the implemented CSV-to-Tableau workflow.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
External Viewer	Explore dashboard insights	USN-1	As a viewer, I want to open the published Tableau workbook so that I can explore global fuel-price and taxation information.	The published workbook opens and the three scenario dashboards are accessible.	High	Release 1
External Viewer	Compare taxation and subsidy policy	USN-2	As an energy-policy viewer, I want to compare average fuel taxes and petrol prices by country or region so that I can inspect patterns across places.	Dashboard 1 displays the taxation, regional petrol-price, tax-versus-price, and subsidy-level views.	High	Release 1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
External Viewer	Review country price rankings	USN-3	As a supply-chain viewer, I want to see the highest and lowest average petrol-price countries so that I can compare potential cost differences.	Dashboard 2 shows top-10 and bottom-10 rankings based on average petrol price, with petrol/diesel comparison.	High	Release 1
External Viewer	Review historical price movement	USN-4	As a supply-chain viewer, I want to review fuel-price trends alongside Brent crude prices so that I can examine how the series move over time.	The dashboard includes the Brent crude and fuel-price trend visualization; the chart supports comparison, not proof of causation.	High	Release 1
External Viewer	Explore income-level patterns	USN-5	As an economics viewer, I want to compare average petrol prices and taxes across income levels so that I can examine group-level differences.	Dashboard 3 includes average petrol price and average tax percentage by income level.	High	Release 1
External Viewer	Filter analytical views	USN-6	As an economics viewer, I want to use the available income-level filter on relevant charts so that I can focus the comparison on a selected group.	The income-level filter is available on the relevant Dashboard 3 views and updates those views as configured.	Medium	Release 1
External Viewer	Interpret categories consistently	USN-7	As a viewer, I want to see category labels and legends so that I can interpret regions, subsidy levels, and other groupings correctly.	Relevant charts show their labels, legends, and titles in the published workbook.	Medium	Release 1
Project Owner / Analyst	Maintain source data and workbook	USN-8	As the project owner, I want to replace or update the CSV source and refresh the workbook when needed so that the analysis can reflect a revised dataset.	A source update is performed by the owner in Tableau and the workbook is republished; no automatic live feed is assumed.	Medium	Future / Maintenance
Project Owner / Analyst	Review data field definitions	USN-9	As the project owner, I want to check the dataset fields and their data types so that dates, prices, tax percentages, and categories are used appropriately.	The source contains the documented fields: date, country, region, income_level, subsidy_level, petrol_usd_liter, diesel_usd_liter, lpg_usd_liter, brent_crude_usd, and tax_percentage.	High	Release 1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
External Viewer	Access workbook without app-specific account features	USN-10	As a viewer, I want to view the publicly published workbook without a project-specific login flow so that I can access the dashboards easily.	The workbook is publicly viewable through its Tableau Public link; no private account or registration feature is claimed.	Medium	Release 1

Project scope and accuracy note: The diagram describes the implemented CSV-to-Tableau workflow at a high level. User stories document capabilities visible in the published workbook and maintenance tasks for the project owner. Release labels are planning labels, not evidence of formal software sprints or a production release process. The published workbook is public, and the project does not claim login, registration, or automated data ingestion.

Customer Journey Map

Date	31 October 2026
Team ID	Individual Project (No Team ID)
Project Name	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Maximum Marks	2 Marks

Customer Journey Map

Map out the customer's experience stage-by-stage, capturing their actions, touchpoints, thoughts, and feelings, along with process ownership and improvement opportunities at each stage.

Phase of Journey	Access & Orient	Explore & Compare	Interpret & Decide
Actions *What does the customer do?*	- Open the published Tableau workbook. - Read the dashboard title and purpose. - Identify the relevant policy question.	- Select a dashboard and compare charts. - Use available filters and legends. - Compare countries, regions or groups.	- Review rankings and trend patterns. - Compare prices with tax or crude-oil trends. - Note findings and limitations.
Touchpoint *What part of the service do they interact with?*	- Tableau Public workbook link - Dashboard title and chart labels - Published workbook landing view	- Three interactive dashboards - Filters, legends and chart tooltips - Country, region, subsidy and income views	- Rankings and time-series charts - Tax/price comparison charts - Export/share or presentation of observations
Customer Thought *What is the customer thinking?*	- What does this workbook cover? - Which dashboard fits my question? - What period and measures are shown?	- Which countries or regions differ? - How do prices vary by subsidy or income level? - Can I compare like-for-like measures?	- What patterns are visible? - Are these associations or proven causes? - What further context is needed?
Customer Feeling *What is the customer feeling?*	Curious; may need brief orientation to understand the dashboard structure.	Engaged when comparisons are clear; may feel uncertain if filters or measures are unfamiliar.	More informed if patterns are clear; cautious about drawing conclusions from descriptive data.

Process Ownership *Who is in the lead on this?*	Project creator / Tableau workbook owner: maintain the published workbook and dataset description.	Dashboard viewer: choose filters and interpret charts; workbook owner maintains labels and chart configuration.	Dashboard viewer: assess patterns; project creator documents scope, definitions and analytical limitations.
Opportunities *How can we improve this stage?*	- Add a short ?how to use? note. - Make date range and units easy to find. - Explain key measures.	- Keep filter labels consistent. - Use clear titles and tooltips. - Avoid overcrowding charts.	- State that trends show association, not causation. - Document dataset coverage and update process. - Validate interpretations with external context.

Project note: This journey map is a design-stage hypothesis based on the published Tableau workbook and intended analytical tasks. It is not based on recorded customer interviews or formal usability testing. Improve it after observing real users.